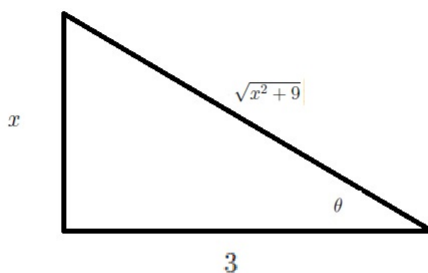


Ayudantía 15 - MAT1610

1. Determine $\underbrace{\int \frac{x^2}{\sqrt{x^2+9}} dx}_I$

Solución:

Considerar la sustitución $x = 3 \tan(\theta)$ o $\frac{x}{3} = \tan(\theta)$ con $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$ entonces $dx = 3 \sec^2(\theta) d\theta$
y $\sqrt{x^2+9} = \sqrt{9 \tan^2(\theta) + 9} = \sqrt{9(\tan^2(\theta) + 1)} = 3|\sec(\theta)| = 3 \sec(\theta)$
 $\sec(\theta) > 0$ porque $\cos(\theta) > 0$ para $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$.



$$\begin{aligned} I &= \int \frac{x^2}{\sqrt{x^2+9}} dx = \int \frac{9 \tan^2(\theta) 3 \sec^2(\theta) d\theta}{\sqrt{9(\tan^2(\theta) + 1)}} \\ &= 9 \underbrace{\int \tan^2(\theta) \sec(\theta) d\theta}_{I_1} \end{aligned}$$

Una forma para determinar I_1 es aplicar integración por partes

$$\begin{aligned}
 I_1 &= \int \tan^2(\theta) \sec(\theta) d\theta \\
 &= \int \underbrace{\tan(\theta)}_u \underbrace{\tan(\theta) \sec(\theta)}_{dv} d\theta \\
 &= \tan(\theta) \sec(\theta) - \int \sec(\theta) \sec^2(\theta) d\theta \\
 &= \tan(\theta) \sec(\theta) - \int \sec(\theta) (\tan^2(\theta) + 1) d\theta \\
 &= \tan(\theta) \sec(\theta) - \underbrace{\int \sec(\theta) \tan^2(\theta) d\theta}_{I_1} - \int \sec(\theta) d\theta \\
 &= \tan(\theta) \sec(\theta) - I_1 - \ln(|\sec(\theta) + \tan(\theta)|)
 \end{aligned}$$

Por lo tanto,

$$2I_1 = \tan(\theta) \sec(\theta) - \ln(|\sec(\theta) + \tan(\theta)|)$$

es decir,

$$I_1 = \frac{\tan(\theta) \sec(\theta) - \ln(|\sec(\theta) + \tan(\theta)|)}{2} + K$$

Así,

$$\begin{aligned}
 I &= 9I_1 \\
 &= \frac{9}{2} (\tan(\theta) \sec(\theta) - \ln(|\sec(\theta) + \tan(\theta)|)) + C \\
 &= \frac{9}{2} \left(\frac{x \sqrt{x^2 + 9}}{3} - \ln \left(\left| \frac{\sqrt{x^2 + 9}}{3} + \frac{x}{3} \right| \right) \right) + C \\
 &= \frac{x \sqrt{x^2 + 9}}{2} - \frac{9}{2} \ln \left(\left| \frac{x + \sqrt{x^2 + 9}}{3} \right| \right) + C
 \end{aligned}$$

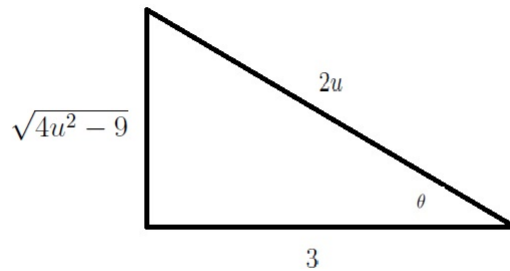
2. Determine $\underbrace{\int \frac{1}{\sqrt{(4x^2 - 24x + 27)^3}} dx}_I$.

Solución:

Note que, $\int \frac{1}{\sqrt{(4x^2 - 24x + 27)^3}} dx = \int \frac{1}{\sqrt{(4(x-3)^2 - 9)^3}} dx$. Entonces, considerando la sustitución $u = x - 3$ se tiene que $du = dx$ y

$$\int \frac{1}{\sqrt{(4x^2 - 24x + 27)^3}} dx = \int \frac{1}{\sqrt{(4u^2 - 9)^3}} du$$

Ahora, haciendo la sustitución $\frac{2}{3}u = \sec(\theta)$, $0 \leq \theta < \frac{\pi}{2}$ o $\pi \leq \theta < \frac{3\pi}{2}$, entonces $du = \frac{3}{2} \sec(\theta) \tan(\theta) d\theta$

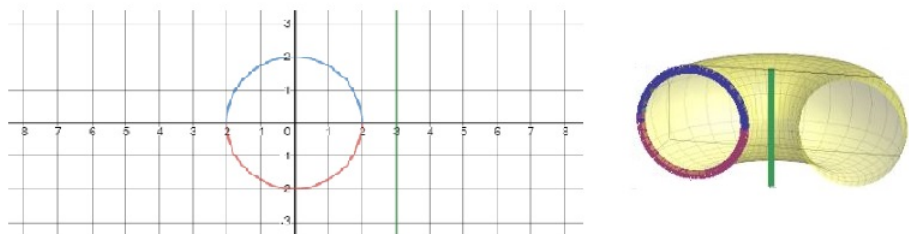


$$\begin{aligned}
I &= \int \frac{1}{\sqrt{(4u^2 - 9)^3}} du \\
&= \int \frac{1}{\sqrt{9^3 \left(\left(\frac{2}{3}u \right)^2 - 1 \right)^3}} du \\
&= \frac{1}{27} \int \frac{1}{\sqrt{(\sec^2(\theta) - 1)^3}} \frac{3}{2} \sec(\theta) \tan(\theta) d\theta \\
&= \frac{1}{18} \int \frac{\sec(\theta) \tan(\theta) d\theta}{\tan^3(\theta)} \\
&= \frac{1}{18} \int \frac{\sec(\theta) d\theta}{\tan^2(\theta)} \\
&= \frac{1}{18} \int \frac{\cos(\theta) d\theta}{\sin^2(\theta)} \\
&= -\frac{1}{18} \frac{1}{\sin(\theta)} + C \\
&= -\frac{1}{18} \frac{2u}{\sqrt{4u^2 - 9}} + C \\
&= -\frac{1}{18} \frac{2(x-3)}{\sqrt{4(x-3)^2 - 9}} + C \\
&= -\frac{1}{9} \frac{(x-3)}{\sqrt{4(x-3)^2 - 9}} + C
\end{aligned}$$

3. Hallar el volumen del toro generado en la rotación del círculo $x^2 + y^2 = 4$ alrededor de la recta $x = 3$.

Solución:

Idea gráfica



$$\begin{aligned}
 V &= \int_{-2}^2 2\pi(3-x) \left(\sqrt{4-x^2} - \left(-\sqrt{4-x^2} \right) \right) dx \\
 &= 4\pi \int_{-2}^2 (3-x)\sqrt{4-x^2} dx \\
 &= 4\pi \left(\underbrace{3 \int_{-2}^2 \sqrt{4-x^2} dx}_{I_1} - \underbrace{\int_{-2}^2 x\sqrt{4-x^2} dx}_{I_2} \right) \\
 &= 4\pi (3I_1 - I_2) \\
 &= 12\pi I_1
 \end{aligned}$$

Note que el integrando de la integral I_2 es impar y el intervalo de integración es simétrico respecto al origen entonces la integral vale cero (Para calcular I_2 considere la sustitución $u = 4 - x^2$, entonces $du = -2x dx$ o $-\frac{du}{2} = x dx$ y si $x = -2$ entonces $u = 0$ y si $x = 2$ entonces $u = 0$).

Para calcular I_1 considere la sustitución $x = 2\sin(\theta)$, con $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$ entonces, $dx = 2\cos(\theta)d\theta$, $\sqrt{4-x^2} = \sqrt{4-4\sin^2(\theta)} = 2|\cos(\theta)| = 2\cos(\theta)$, $\theta = \arcsin\left(\frac{x}{2}\right)$ y si $x = 0$ entonces $\theta = \arcsin\left(\frac{0}{2}\right) = 0$ y si $x = 2$ entonces $\theta = \arcsin\left(\frac{2}{2}\right) = \arcsin(1) = \frac{\pi}{2}$.

Por lo tanto,

$$\begin{aligned} I_1 &= \int_{-2}^2 \sqrt{4-x^2} dx \\ &= 2 \int_0^2 \sqrt{4-x^2} dx \\ &= 2 \int_0^{\frac{\pi}{2}} 2 \cos(\theta) 2 \cos(\theta) d\theta \\ &= 8 \int_0^{\frac{\pi}{2}} \cos^2(\theta) d\theta \\ &= 8 \int_0^{\frac{\pi}{2}} \frac{1 + \cos(2\theta)}{2} d\theta \\ &= 4 \int_0^{\frac{\pi}{2}} (1 + \cos(2\theta)) d\theta \\ &= 4 \left(\theta + \frac{\sin(2\theta)}{2} \right) \Big|_0^{\frac{\pi}{2}} \\ &= 4\theta + 2\sin(2\theta) \Big|_0^{\frac{\pi}{2}} \\ &= 2\pi \end{aligned}$$

Así $V = 12\pi \cdot 2\pi = 24\pi^2$ unidades de volumen.

4. Considere la sustitución $u = \sqrt{\frac{1-x}{1+x}}$ en

$$\underbrace{\int \frac{1}{x^2} \sqrt{\frac{1-x}{1+x}} dx}_I$$

y resuelva.

Solución:

Sea $u = \sqrt{\frac{1-x}{1+x}}$ entonces,

$$du = \frac{1}{2\sqrt{\frac{1-x}{1+x}}} \left(\frac{1-x}{1+x}\right)' dx = \frac{1}{2\sqrt{\frac{1-x}{1+x}}} \left(-1 + \frac{2}{1+x}\right)' dx = \frac{1}{\sqrt{\frac{1-x}{1+x}}} \left(-\frac{1}{(1+x)^2}\right) dx$$

Note que $u = \sqrt{\frac{1-x}{1+x}} = \sqrt{-1 + \frac{2}{1+x}}$, entonces $u^2 = -1 + \frac{2}{1+x}$, $\frac{u^2+1}{2} = \frac{1}{1+x}$ ó $\frac{(u^2+1)^2}{4} = \frac{1}{(1+x)^2}$,
 $x = \frac{2}{u^2+1} - 1 = \frac{1-u^2}{u^2+1}$ y $x^2 = \frac{(1-u^2)^2}{(u^2+1)^2}$. Por lo tanto,

$$du = -\frac{1}{u} \frac{(u^2+1)^2}{4} dx \text{ o } dx = -\frac{4u}{(u^2+1)^2} du \text{ y}$$

$$\begin{aligned} I &= - \int u \frac{(u^2+1)^2}{(1-u^2)^2} \frac{4u}{(u^2+1)^2} du \\ &= -4 \int \frac{u^2}{(1-u^2)^2} du \\ &= -4 \int \frac{u^2}{(1-u)^2(1+u)^2} du \\ &= -4 \left[\int \frac{A}{1-u} du + \int \frac{B}{(1-u)^2} du + \int \frac{C}{1+u} du + \int \frac{D}{(1+u)^2} du \right] \\ &= -4 \left[-A \ln(|1-u|) + B \frac{1}{1-u} + C \ln(|1+u|) - D \frac{1}{1+u} \right] + K \\ &= -4 \left[\frac{1}{4} \ln(|1-u|) + \frac{1}{4(1-u)} - \frac{1}{4} \ln(|1+u|) - \frac{1}{4(1+u)} \right] + K \\ &= -4 \left[\frac{1}{4} \ln(|1-u|) + \frac{1}{4(1-u)} - \frac{1}{4} \ln(|1+u|) - \frac{1}{4(1+u)} \right] + K \\ &= -\ln(|1-u|) - \frac{1}{1-u} + \ln(|1+u|) + \frac{1}{1+u} + K \\ &= -\ln(|1-u|) + \frac{1}{u-1} + \ln(|1+u|) + \frac{1}{1+u} + K \\ &= -\ln \left(\left| 1 - \sqrt{\frac{1-x}{1+x}} \right| \right) + \frac{1}{\sqrt{\frac{1-x}{1+x}} - 1} + \ln \left(\left| 1 + \sqrt{\frac{1-x}{1+x}} \right| \right) + \frac{1}{1 + \sqrt{\frac{1-x}{1+x}}} + K \end{aligned}$$

El valor de las constantes A, B, C y D se obtiene de plantear que para todo u :

$$u^2 = A(1-u)(1+u)^2 + B(1+u)^2 + C(1-u)^2(1+u) + D(1-u)^2$$

o

$$u^2 = A(1-u^2)(1+u) + B(1+u)^2 + C(1-u)(1-u^2) + D(1-u)^2$$

o

$$u^2 = A(1 + u - u^2 - u^3) + B(1 + 2u + u^2) + C(1 - u - u^2 + u^3) + D(1 - 2u + u^2)$$

es decir,

$$u^2 = (-A + C)u^3 + (-A + B - C + D)u^2 + (A + 2B - C - 2D)u + (A + B + C + D)$$

lo cual implica que:

$$-A + C = 0$$

$$-A + B - C + D = 1$$

$$A + 2B - C - 2D = 0$$

$$A + B + C + D = 0$$

De la primera ecuación se tiene que $A = C$, sustituyendo en la tercera se obtiene que $B - D = 0$, es decir, $B = D$. Usando dichas relaciones en la cuarta ecuación se obtiene que $2A + 2B = 0$ ó $B = -A$ y por ello, $B = -C$

Ahora sustituyendo en la segunda ecuación se tiene que $-A - A - A - A = 1$, esto es, $-4A = 1$ y en consecuencia, $A = -\frac{1}{4}$, $C = -\frac{1}{4}$, $B = \frac{1}{4} = D$